$$\mathbf{u} \odot \mathbf{v} = \frac{\mathbf{q}+\mathbf{d}}{\mathbf{K}\mathbf{v}} \odot \frac{\mathbf{q}-\mathbf{s}+\mathbf{d}}{(\mathbf{K})^{\top}\mathbf{u}} = \mathbf{1}$$

Due to the constraint on $\mathbf{q}$:

$$\mathbf{q}^{*} = \min\left(\frac{\mathbf{s}}{2} + \left((\mathbf{K}\mathbf{v}) \odot (\mathbf{K}^{\top}\mathbf{u}) + \frac{\mathbf{s}^{2}}{4}\right)^{\frac{1}{2}} - \mathbf{d}, \mathbf{r}\right) \tag{31}$$

Due to $\mathbf{P} \leq \mathbf{S}$,we can get that :

$$\mathbf{P} = \mathrm{Diag}(\mathbf{u})e^{(-\mathbf{D}+\mathbf{H})/\epsilon}\mathrm{Diag}(\mathbf{v}) \leq \mathbf{S} \tag{32}$$

Thus: $\mathbf{H} \leq \mathbf{D} + \epsilon \cdot \log\left(\mathrm{Diag}(\mathbf{u}^{-1})\mathbf{S}\mathrm{Diag}(\mathbf{v}^{-1})\right)$ Thus we can get that :

$$\mathbf{H} = \min\left(\mathbf{D} + \epsilon \cdot \log\left(\mathrm{Diag}(\mathbf{u}^{-1})\mathbf{S}\mathrm{Diag}(\mathbf{v}^{-1})\right), \mathbf{0}\right) \tag{33}$$

D DETAILS ABOUT EXPERIMENTS

D.1 DETAILS ABOUT PARAMETERS

Table 4 show the parameters of MCF experiments at Table 1 in main text ,including the Batch size, Regularization coefficients, Split of the node set, $\mathbf{d}$ (Virtual-flow), $\mathbf{Node\_C}$ (Capacity for flows at nodes), $\mathbf{Edge\_C}$ (Capacity for edges)), and $\mathbf{err}$ (Convergence threshold of the iterations).

Table 4: The parameters for experiments on synthetic dataset

Instance	Instance_Num	ϵ	Node_Split	d0	Edge_C	Node_C	Convergence threshold
Node_500_w/o constraint	64	5e-4	50_400_50	1e-3	-	-	1e-6
Node_500_w/edge constraint	256	1e-4	50_400_50	1e-4	0.05	-	1e-5
Node_500_w/node constraint	256	5e-4	50_400_50	1e-4	-	0.1	1e-5
Node_500_w/ edge+node constraints	256	1e-4	50_400_50	1e-4	0.1	0.5	1e-5
Node_5k_w/o constraint	16	5e-4	500_4000_500	1e-4	-	-	1e-6
Node_5k_w/edge	16	5e-4	500_4000_500	1e-4	0.05	-	1e-5
Node_5k_w/node	8	1e-3	500_4000_500	1e-4	-	0.1	1e-5
Node_5k_w/ edge+node constraints	8	5e-4	500_4000_500	1e-4	0.1	0.5	1e-5
Node_10k_w/o constraint	4	5e-4	1000_8000_1000	1e-6	-	-	1e-6
Node_10k_w/ edge constraints	2	5e-4	1000_8000_1000	1e-4	0.05	-	1e-5
Node_10k_w/ node constraints	2	5e-4	1000_8000_1000	1e-4	-	0.1	1e-5
Node_10k_w/ edge+node constraints	2	5e-4	1000_8000_1000	1e-4	0.1	0.5	1e-5

Similarly, Table 5 show the parameters of MCF experiments at Table 2 in main text, including the Batch size, Regularization coefficients, Split of the node set, $\mathbf{d}$ (Virtual-flow), $\mathbf{Cap\_Range}$ (Capacity range for NETGEN to randomly generate), $\mathbf{Arc\_Num}$ (Totak arc number)), and $\mathbf{err}$ (Convergence threshold of the iterations).

The Convergence threshold $\mathbf{err}$ is defined as the average differance of the marginals:

$$err = \frac{\sum_{i=1}^{\text{batch\_size}} \left\|\mathbf{P}_i\mathbf{1}_n - \mathbf{P}_i^{\top}\mathbf{1}_m - \mathbf{s}_i\right\|}{\text{batch\_size}}$$

Table 5: The parameters for experiments on NETGEN dataset

Instance	Instance_Num	ϵ	Node_Split	d0	Cap_Range	Arc_Num	err
Node_100_NETGEN	128	1e-3	10_80_10	1e-4	[0.5,1]	800	1e-5
Node_500_NETGEN	256	5e-4	50_400_50	1e-4	[0.5,1]	64 k	1e-4
Node_1k_NETGEN	128	5e-4	100_800_100	1e-4	[0.5,1]	80 k	1e-4
Node_5k_NETGEN	8	5e-4	500_4000_500	1e-4	[0.5,1]	10000 k	1e-4

Table 7 shows the Sparsity degree of the Table 1,including the Cost Matrix, Guroobi and EOFT.The results shows that the Uniform-dataset is sparse graph, and both Gurobi and EOFT have obtained nearly equally sparse solutions across various constraint scenarios and problem sizes. The arc number of NETGEN dataset is about $O(n^2)$ on 500/1k/5k instances, which means they are Dense graph.